

‘The Materials of Science, the Ideas of Science, and the Poetry of Science’: Warren McCulloch and Jerry Lettvin

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This essay explores the relationship between neurophysiologist and cybernetician Warren S. McCulloch (1898–1969) and one of his most important collaborators, neurophysiologist Jerome Y. Lettvin (1920–2011). Lettvin met McCulloch in 1941 as a medical student at the University of Illinois at Chicago, where McCulloch was associate professor of psychiatry. Lettvin recalled that it was McCulloch’s influence that led him to study the nervous system; however, theirs was a distinct sort of relationship between mentor and student. According to Lettvin, McCulloch treated young scientists not as children but as if they were experts in any field they happened to be discussing. Both shared an irreverent spirit and ultimately, McCulloch instilled in Lettvin a penchant for theoretical investigations of the nervous system and a humanistic approach to the sciences. By tracing McCulloch’s and Lettvin’s relationship from their initial meeting in 1941 through to their famous collaborations at MIT’s Research Laboratory of Electronics in the 1950s and 1960s, this article highlights the complexities of mentor–student relationships in twentieth-century American science, the personal and intellectual alliances that defined work in cybernetics, and the ‘transdisciplinary’ sorts of practices that characterized cybernetic studies of the brain.

KEYWORDS cybernetics, Jerome Y. Lettvin, Warren S. McCulloch, mentor relationships

Introduction

Towards the end of an interview with Percy Saltzman on the CBC program ‘The Day It Is’ during the late 1960s, American neurophysiologist and cybernetician Warren McCulloch (1898–1969) was asked for his thoughts on the legacy of his work. Smiling, he replied ‘Oh, that’s very easy. It’s the youngsters who work with me. I’ve had a hundred and twenty younger — more than a hundred and twenty — younger collaborators. Forty of these are very good scientists in their own right today’.¹ Indeed, after McCulloch’s death in September 1969 of a coronary seizure, his friend and fellow

neurophysiologist Ralph W. Gerard wrote that McCulloch ‘loved and attracted young people. . . and I suspect that he would relish more his affinity with and influence on the young than he would his more formal and formidable professional contributions’ (Gerard 1970, 344).

Taking it as a given that a scientist’s legacy and a history of his or her influence must take into account more than scientific results and theories, this essay explores the relationship between Warren McCulloch and one of his most important students and closest collaborators, neurophysiologist Jerome Y. Lettvin (1920–2011). Although never a core member of the group of cyberneticians that included McCulloch, Walter Pitts (1924–1969), and arguably the most famous of the group, MIT mathematician Norbert Wiener (1894–1964), Lettvin made key contributions to our theoretical and experimental understanding of the central nervous system and ultimately became a dynamic figure at MIT during the 1960s, 1970s, and 1980s. Lettvin met McCulloch in 1941 as a medical student at the University of Illinois at Chicago, where McCulloch was associate professor of psychiatry and head of the Research Laboratory at the newly opened Illinois Neuropsychiatric Institute (INI). Lettvin recalled that it had been McCulloch’s influence that had initially led him to study the nervous system; however, their relationship was not a typical one between mentor and student. McCulloch, Lettvin recalled, treated young scientists not as children but as if they were experts in any field they happened to be discussing (Lettvin 1989a, 337). Invited to stay at McCulloch’s home in Chicago for a year in 1942 (along with McCulloch’s better-known collaborator, Walter Pitts),² Lettvin recalled that it was this experience that influenced him more than any other in how he treated ‘the materials of science, the ideas of science, and the poetry of science’ (Lettvin 1989a, 337).

These three features highlight the sorts of scientific practices that were characteristic of both McCulloch and Lettvin. I take the ‘materials of science’ to mean the laboratory practices that defined neurophysiology, the disciplinary ‘home’ of both men. Both McCulloch and Lettvin were trained as experimentalists and did extensive laboratory work on the central nervous system, earning their keep in psychiatric hospitals. The ‘ideas of science’ relates to a distinct but related form of scientific practice — McCulloch’s and Lettvin’s framing of particular questions about the brain and views on the methodology appropriate for studying the brain. Especially as they each matured as scientists, both McCulloch and Lettvin were frustrated at what they saw as an overemphasis on the empirical study of the nervous system, and wanted more room for theoretical speculation. In addition, both pursued more humanistic approaches to the sciences and frequently offered philosophical reflections; for his part, McCulloch was concerned with scientific philosophy and history of philosophy, Lettvin with the history of science.³ In terms of the ‘poetry of science’, McCulloch and Lettvin were both poets, shared an irreverent, carefree spirit, and delighted in wordplay. The extent to which McCulloch ‘instilled’ all of these traits in Lettvin is an unanswerable question — however, a closer look at McCulloch’s mentoring style suggests a complex picture. Although McCulloch was certainly a mentor to Lettvin, he had a particular style of mentoring.

Recently, historians have pointed to the complex ways in which scientific roles and personae play in the sciences and in scientific pedagogy (e.g. Kaiser 2005). For example, as Robert Kohler has shown in his study of geneticist Thomas Hunt Morgan and the moral relationships that defined the ‘fly group’ at Columbia and Caltech, mentor–student relationships are about more than

what happens inside the laboratory or the classroom (Kohler 1994). In his analysis of the organization of research groups in biochemistry in nineteenth-century German universities, Joseph S. Fruton examined the role of mentors and mentoring practices in research schools (Fruton 1990). He contrasted two styles of organization in these groups, with one style having a more authoritarian, 'quasi-military' mentoring style, controlling most aspects of students' work, and the other having a much more liberal, egalitarian approach to students, with the mentor encouraging them to be independent researchers, treating them less as students than as colleagues (Fruton 1990, Chapter 1). Although the group that formed around McCulloch was not a research school in the traditional sense (cf. Geison and Holmes 1993), he was certainly an influence and inspiration to numerous figures in neurophysiology, cybernetics, and artificial intelligence.⁴ And, as the story of his relationship with Lettvin shows, McCulloch's style as a mentor certainly aligns with the more liberal type of mentor; his style was much more 'laissez-faire' than authoritarian.⁵ As a father, McCulloch had encouraged his children to be self-reliant,⁶ and this translated to his role as a mentor: although he was a guide to them intellectually and scientifically, he always encouraged their independence and self-reliance.

In tracing McCulloch's and Lettvin's personal and intellectual relationship from their initial meeting in 1941 through to their famous collaborations at MIT's Research Laboratory of Electronics in the 1950s and 1960s, this essay aims to add to our understanding of the complexities that defined mentor-student relationships in twentieth-century American science. Through an examination of Lettvin's own recollections of McCulloch, recollections of Lettvin by his peers, their correspondence, and their collaborative efforts in understanding the nervous system, this story also highlights a key example of the kinds of personal and intellectual alliances that defined work in cybernetics and the extent to which McCulloch's legacy lies in his social and intellectual generosity. In light of McCulloch's and Lettvin's mingling of humanistic and scientific pursuits, ultimately their story can tell us much about the kinds of practices that defined 'cybernetic' studies of the brain, which were practised in a similar transdisciplinary spirit.

Mentor and student: Chicago and the Illinois Neuropsychiatric Institute

Jerome Ysroael Lettvin was born in Chicago on February 23, 1920 to Ukrainian immigrants. Growing up in what he called an 'enchanted ghetto, Humboldt Park' (Lettvin 1998a, 224), Lettvin had early aspirations to be a poet, but following high school, his mother made it clear that Lettvin would go into medicine or 'all funds would stop' (Lettvin, 1998b, 1). From 1936 to 1938, Lettvin attended the Lewis Institute, a junior college that eventually became the Illinois Institute of Technology. Lettvin attended the University of Chicago from 1938 to 1939 and was instructed in brain anatomy by neuroanatomist Gerhardt von Bonin.

Lettvin then entered medical school at the University of Illinois, where he graduated with a BS in 1942 and an MD in 1943. During Lettvin's third year of medical school, von Bonin took him over to the Illinois Neuropsychiatric Institute (INI) at the University of Illinois to meet Warren McCulloch. Lettvin recalled the meeting:

When I was in high school, I had read Buckle's history of the Scotch mind in the seventeenth century. It gave me nightmares for weeks afterward. Warren would be the first Scotchman I ever knew, and truly, I had a frisson going to see him. This was not lessened when I met him. He sported no beard then, but certainly had the most piercing eyes I ever encountered. However, in very little time, he, acting most friendly, dispelled my nervousness (Lettvin 1998a, 227).

Lettvin ultimately joined the INI as a student, and under McCulloch's influence, Lettvin became 'absolutely committed' to working on the nervous system.

Lettvin soon learned that McCulloch took a keen interest in personally supporting the young people and students that he happened to encounter. As McCulloch's wife Rook Metzger McCulloch recalled, he had an 'ease of communication' with every age and every sort of person' (McCulloch 1989, 2). McCulloch and Rook were both very generous, and notorious for taking cash-strapped students into their home in Hinsdale, Illinois, a rural suburb on the outskirts of Chicago. Early in 1942, both Lettvin's friend Walter Pitts (at the time sixteen and homeless, but would become a core member of the cybernetics group) and Lettvin (twenty-two and under 'intense' supervision at home) moved in to the McCulloch home (Conway and Siegelman 2005, 139). The McCulloch household was a free-spirited environment, always crowded with houseguests coming and going, many from McCulloch's group at the University of Illinois. As Barbara Wiener, who stayed with the McCullochs for some time during 1947, recalled: 'The laird of the castle appeared and disappeared at will, talking grandly and holding court. The household was proudly radical. The phonograph in the living room kept spinning out songs of the Spanish Civil War and union songs to which everyone knew the words but me' (Raisbeck in Conway and Siegelman 2005, 202).

Lettvin has described McCulloch as having distinct approach to teaching that encouraged students to think for themselves and to think critically. In Lettvin's words, McCulloch taught students 'the art of criticism, that is, how to take all possible avenues toward breaking up notions and resynthesizing them' (Lettvin 1989a, 333). McCulloch's style of communication was very literary, and his teaching of problems in neurophysiology was always peppered with personal anecdotes, literary references, and allusions to fanciful stories. Lettvin and many other students found this inspiring, although it frustrated and daunted some of McCulloch's more seasoned listeners (Lettvin 1989b, 335). At the INI, where many of McCulloch's colleagues had emphasized the pursuit of 'data and only data', McCulloch's emphasis on ideas and theoretical questions captivated his students. Unafraid of being wrong, McCulloch taught Lettvin that 'wrong guesses and wrong theories may be much more important than the acquisition of new data', and 'continually practiced theorizing and taught his students the practice of theorizing as a way of designing proper experiments' (Lettvin, 1989b, 336).

By August 1942, when Lettvin was about to begin his last year of medical studies at Illinois, McCulloch indicated that he'd be willing to write recommendation letters for him, since he had 'a reasonably long and close association' with Lettvin and expected him to live up to his early promise.⁷ Lettvin had applied for a position as house officer at Boston City Hospital. McCulloch wrote to the director that he'd known Lettvin for about a year in the Research Laboratory of the INI, and described Lettvin's interests in neurology and his 'outstanding ability'. McCulloch wrote of Lettvin's superb manual dexterity and related that he was a 'happy, friendly boy, thoroughly healthy and hard working'.⁸

The fall of 1942 saw Lettvin effectively drop out of medical school: he kept up with his medical school examinations, but did not regularly attend classes. He was still affiliated with the INI, but owing to the shortages of medical personnel as a result of the Second World War, he responded to the need for doctors, nurses, and interns on the clinical floors of the INI and took on whatever role was necessary (Lettvin 1998b, 3). This clinical experience left Lettvin much more passionate about neurology, and the next few years saw him gaining more experience in clinical settings. He got accepted into the Harvard Neurological Service and spent part of his time between 1943 and 1944 working in neurology and part of his time working on medical wards, interning at Boston City Hospital (BCH), working with New Zealand-born neurologist Derek Denny-Brown. Shortly after his arrival in Boston, Lettvin wrote to McCulloch to say that he'd arrived safely, and aside from some professional gossip, it was a very warm affectionate letter that attests to their intimacy:

People aren't cold out here, save for a few — and I have met a fine Hungarian surgeon with whom I play Bach on the flageolet — for he has that vice too. Why do Bostonians pronounce 'room' as if it were something to drink?...Love, Jerry.⁹

McCulloch also introduced Lettvin to Harvard physiologist Arturo Rosenblueth by correspondence in March 1943, and enthusiastically described Lettvin as having 'more brains than any man in his class, and he is spoiled rotten in the matter of highly intelligent friends. His closest companion is a youngster named Pitts...'.¹⁰ McCulloch urged Rosenblueth to welcome Lettvin, and said they would find him 'gay, intelligent, and really friendly. Don't hesitate to spank him if he gets too boisterous...'.¹¹

From January 1944 to 1946, Lettvin was in the US Army. In 1944, he reported to New York's Bellevue Hospital for training in neuropsychiatry (working with McCulloch's former colleague S. Bernard Wortis), then to Crile General Hospital near Cleveland and ultimately as first Lieutenant and Head of Neuropsychiatry at the 237th General Hospital in France until 1946, serving principally as a psychiatrist but using neurophysiology to approach neurological problems. As Pitts wrote to McCulloch in September 1944,

the army is not making a man out of [Lettvin]; he is annoyed with the vast quantity of bourgeois doctors who go about strutting in their officer's uniforms, being manly and virile in a boy-scientist sort of way... Jerry, despite the best efforts of his family, is incurably proletarian; and not even an Army commission can ever make him [a] respectable, admirable man!¹²

Lettvin returned to the US in 1946, working at various other hospitals, and was discharged from the army in September. Upon his return, Lettvin began experiencing a crisis of identity. He recalled that his former teacher at BCH, Denny-Brown, was 'cold' to him. When Lettvin asked him what was wrong, Denny-Brown replied, 'You became a psychiatrist'. Lettvin answered: 'I didn't become a psychiatrist. You know I'm a neurologist. They forced me'. Denny-Brown said that this was no excuse.¹³ Lettvin recalled that he did want to return to working in neurology, but the label 'psychiatrist' was hard to shake.

March 1946 saw the first Macy Foundation-sponsored gathering in New York City of the interdisciplinary group of scientists that became known as 'the cybernetics group' (Heims 1993). Perhaps influenced by Pitts and this burgeoning field, that spring Lettvin wrote to Wiener at MIT with plans for

brushing up on mathematics. Pitts, who was Lettvin's best friend, was at MIT at the time working with Wiener, splitting his time, as Wiener was, between Cambridge and Mexico (to which Rosenblueth had returned). Lettvin wanted to study the physical sciences and mathematics, which he felt was necessary in neurophysiology and psychiatry 'in the light of the new work done by Mr Pitts, Dr McCulloch, and you'.¹⁴ Lettvin cited his years of training in medicine, neurophysiology, and neuropsychiatry, but told Wiener that his

unofficial training, however, is of rather more importance. I have been connected with Dr McCulloch and his laboratory on and off for the past five years and have been in constant communication with Mr Pitts for the past six years. From these men I have learned in a general way what the great problems are in neurophysiology, especially insofar as treatment of nerve nets is concerned, and also have been following their work in such problems. But to do any original research or to be able to tie in with the work at its present stage I need a good amount of sophistication in mathematics.¹⁵

In the fall of 1946, Lettvin came to MIT as a special student and worked with Wiener while pursuing his 'day job' as a Veteran's Administration psychiatrist.¹⁶ Pitts, Lettvin, and Oliver Selfridge shared a flat at 139 Beacon Street in Boston, and they had also befriended MIT historian Giorgio de Santillana around this time. Lettvin recalled this year he spent at MIT gave him a 'freshman introduction to science' and that he learned differential geometry with Dirk Struik, among other things (Lettvin 1998a, 231).

By the spring of 1947, a falling out had taken place between Wiener, Pitts, McCulloch, and Lettvin over a manuscript of Wiener's that Wiener believed had been misplaced by Pitts. Wiener also accused Pitts, Lettvin, and others of participating in a conspiracy of silence about it.¹⁷ Pitts at the time was in Mexico with Rosenblueth; Wiener wrote to McCulloch asking for McCulloch's knowledge of the situation, suspecting McCulloch's involvement.¹⁸ McCulloch's response is telling. Wiener was a true 'father figure' to Pitts, writing to McCulloch with a list of things to do to keep Pitts in line and focused on his work. McCulloch, by contrast, was less staunchly 'disciplinarian'. McCulloch admitted to Wiener that he'd even suspected Lettvin had lost one of his own manuscripts, but did not hold it against him; partly because he knew 'of old that Jerry loses things'.¹⁹ He went on to describe some of Lettvin's scientific ideas as erroneous, but again went easy on him:

Jerry's brain produces many notions, of which some are excellent; but someone will always have to stand over him to see to it that the nonsensical ones are not let loose on an unsuspecting world. Hold me ready to inspect any of his products to nip the nonsense in the bud, knowing that in the end, no matter how petulant he may be at the moment, he is basically an affectionate child who will, in the long run, be grateful for the lash.²⁰

Clearly, McCulloch saw his role as a mentor as someone to guide Lettvin scientifically and intellectually and discipline his thinking. However, this did not extend to Lettvin's character. As McCulloch continued to Wiener:

Per contra, there is absolutely no use in my saying anything to him about manuscripts. It might make him suffer at the moment, it would never prevent his losing the next, and he would in all kindness of heart, to spare our feelings or forestall our ill opinions of anyone else, tell us only what he thought was good for us to know.²¹

McCulloch explained the entire situation to Wiener, and Lettvin's role in the affair, which rather than being a conspiracy, was a case of miscommunication (or lack of communication) on the part of Pitts and Lettvin. McCulloch defended both Pitts and Lettvin and suggested the following to Wiener, who had angrily deployed letters about the manuscript loss to Rosenblueth and McCulloch, but not directly to either Pitts or Lettvin:

...here's one piece of advice from a friend with grown sons. Write to these fellows even if it is only to call them turtle's eggs, do not write about them. I have shared your letter fully — and my answer — with Walter who is devoted to you. . . As for Jerry, spank him thoroughly for a poor hypothesis and an ill-defined notion of what's good for you, not for the delayed manuscript.²²

Wiener ultimately forgave Pitts, but not Lettvin. McCulloch wrote a comforting letter to Lettvin that reflects his role as both a personal and scientific mentor. He explained his perspective on Wiener's psychology to Lettvin, and advised him in the future not to be silent to Wiener even if the news is not good. McCulloch reassured Lettvin that his good intentions would not allow Wiener to hold anything against him 'unduly'.²³ McCulloch had even more advice as an intellectual mentor:

Think out and chew over scientific hunches with someone really competent in the details of the field, and somebody like Walter or me who will tear them down from the theorists point of view while we take the trouble to check their hypothetical beliefs in the apparent facts of physiology or what not before you let them escape into the world. . . No matter how engaging such a notion may be. . . let one of us be your mentors. . .²⁴

After the falling out with Wiener, Lettvin returned to McCulloch's laboratory at the INI in April 1947 and worked there until July 1947. McCulloch gave Lettvin 'the run of the laboratory' (Lettvin 1998a, 231). No stranger himself to moving with ease between disciplinary boundaries, McCulloch also helped Lettvin sort his identity issues out. McCulloch told Lettvin: 'Look, the easiest way to get out of this bind is to tell people that you're a physiologist, and see who bites'. Lettvin replied 'I'm not a physiologist', and McCulloch said 'You'll become one'. Lettvin recalled fondly, 'He was a sweetheart'. (Lettvin 1998b, 6–7).

McCulloch described Lettvin in another recommendation letter to neurophysiologist David P. C. Lloyd of the Rockefeller Institute for Medical Research as 'without doubt, the brightest student I have ever had, a highly competent neurologist and one of the rare psychiatrists who really wants to know what is going on in the brains of patients'.²⁵ At this point, Lettvin had had some laboratory experience with use of the stereotaxic instrument,²⁶ but was still in need of more training. He had with McCulloch 'run less than twenty animals' but does his own surgery with good results. McCulloch hoped that Lloyd could help Lettvin, adding that Lloyd would find Lettvin 'full of the exuberance of youth and not yet disciplined in laboratory procedures, but an apt pupil and an excellent scholar. . .'.²⁷ Although McCulloch clearly was playing a mentoring role for Lettvin here, he still presented Lettvin as an independent worker. Their joint research on mapping inhibitory pathways in the central nervous system, presented before the American Physiological Society, in McCulloch's view was entirely Lettvin's own — McCulloch had merely supervised it.²⁸ Further, their joint review of somatic functions of the central nervous system, published in 1948, was, according to McCulloch, chiefly Lettvin's work; McCulloch had simply provided a broad outline.²⁹

Collaborators: MIT's research laboratory of electronics

Following work with psychologist G. R. Wendt at the University of Rochester on a federal contract, Lettvin returned to the INI in 1948 to work with McCulloch on synaptic transmission, a project supported by funds from the Office of Naval Research.³⁰ At this time McCulloch described him as 'the most brilliant student I have yet found at the University of Illinois' and a 'close personal friend'.³¹ During his work at Manteno State Hospital, about an hour's drive outside of Chicago, between 1948 and 1951, Lettvin had set up a neurophysiology laboratory for investigating the site of action of drugs. Although he had been hired as a psychiatrist here, he accepted the position on the condition that he'd be able to set up a neurophysiology laboratory.³² At Manteno, Lettvin transformed himself into a neurophysiologist. He learned to build his own apparatus, such as amplifiers and oscilloscopes (Lettvin 1998b, 8), and collaborated from time to time with Patrick D. Wall (1925–2001), who at the time was an assistant professor in anatomy at the University of Chicago. While at Manteno, Lettvin was still in close contact with McCulloch, and was very productive,³³ coauthoring a paper with McCulloch and Pitts on central inhibition and excitation in the nervous system.³⁴ After three and a half years at Manteno, Lettvin came back to MIT in 1951, on firmer footing intellectually and professionally, to the Research Laboratory of Electronics.

The Research Laboratory of Electronics (RLE) was an interdisciplinary and interdepartmental unit at MIT that exemplified the alliances that were emerging between the US military, industry, and academic institutions in the Cold War period (Leslie 1993, Chapter 1). Founded in 1946 to transform the wartime Radiation Laboratory, with federal funds from the Joint Services Electronics Program, the aim was to forge a new disciplinary structure for research at MIT, with faculty members drawn primarily from physics and electrical engineering.³⁵ The RLE was devoted to basic research tied to military interests, with the Department of Defense responsible for nearly all of the RLE budget during the 1950s (Leslie 1993, p. 28). Focused broadly on electronics and communications sciences, at the outset, every staff member had been engaged in one way or another in government research during World War II. By the 1950s, the RLE was not defined by a singular scientific approach or domain. Work here included projects that reflected the World War II origins of the lab: electronics, spectroscopy, acoustics, and plasma dynamics.

In 1951, Jerry Wiesner, then head of the RLE, invited McCulloch, Lettvin, and Wall to the RLE as research associates.³⁶ Pitts and Lettvin were delighted to be reunited and at the prospect of all of them working together.³⁷ McCulloch left his position as full professor of psychiatry at Illinois. The group set up a laboratory in Building 20, a temporary structure erected during the Second World War and at the time connected to the RLE. By 1962, about half of the work at the RLE was devoted to communications sciences and engineering, which included statistical communication theory, artificial intelligence, linguistics, neurophysiology, and neurology.³⁸ Funding for McCulloch's, Lettvin's, and Wall's work came from the US National Institutes of Health, the Office of Naval Research, the Air Force, and Bell Laboratories.

Soon after the arrival of McCulloch and his group, Wiener expressed anxiety to MIT president James R. Killian over the 'status' of cybernetics at MIT, and dismay at Pitts and Lettvin, particularly Pitts's 'delinquency'.³⁹ Further, McCulloch was a target as well for coming to the RLE to do physiology with government support — Wiener for his part had been proud

of not accepting 'one cent of government money on behalf of cybernetics or any other project since the last war'.⁴⁰ Wiener described McCulloch as a 'picturesque and swashbuckling figure in science whose attractiveness is considerably in advance of his reliability'.⁴¹ And Wiener clearly saw Lettvin's and Pitts's loyalties lying with McCulloch, who, in his view, 'bought his influence with him by pandering to their desire to be free of all restriction and control'.⁴² Wiener ultimately closed off all relations with the RLE and by this time he had also done so with the Macy group,⁴³ in part due to what he saw as McCulloch's 'expansive personality' and his efforts to 'aggrandize his role in cybernetics'. Over the following years, Wiener continued to distance himself not only from the Macy group but from McCulloch and the 'problem boys'.⁴⁴

The institutional structure at the RLE, as well as the life stages McCulloch and Lettvin found themselves in (McCulloch well into his fifties, Lettvin in his thirties) transformed Lettvin and McCulloch as individual scientists, and ultimately, of course, their relationship. Soon after his arrival at MIT, McCulloch became less interested in experimenting and more interested in theoretical pursuits, and he essentially retreated from laboratory work: Lettvin claims that after 1952, McCulloch never touched physiological experiment again (Lettvin 1989b, 522). McCulloch, Lettvin recalled, became 'seduced into what can be done theoretically with nerve networks', and an interesting group convened around McCulloch, working in this same spirit. A glance at McCulloch's publications from his post-MIT period attests to this: of the over one hundred publications authored or coauthored by McCulloch following his arrival at MIT in 1951, over sixty are from a theoretical or philosophical perspective (many elaborating a cybernetic approach to the brain) and of the publications based on more experimental work, most are coauthored by McCulloch with his collaborators at the RLE: Jerry Lettvin, Pat Wall, and Walter Pitts. The experimental publications that had McCulloch's name attached — of which there are at least ten — were probably out of respect.⁴⁵

One of the most famous products of these collaborations was the 1959 publication 'What the Frog's Eye Tells the Frog's Brain', supported by federal grants from the US Army Signal Corps, the Office of Scientific Research, and the Office of Naval Research (Lettvin *et al.* 1959). One of the most cited papers in the Science Citation Index, the paper demonstrated the existence of 'feature detectors' in the optic nerve of the frog, specific neurons that respond to specific features in a visual stimulus — movement, contrast, curvature, and dimming. Although the author list for the paper included McCulloch, the paper was actually the product of a collaboration between Lettvin and Chilean biologist Humberto Maturana (b. 1928). Maturana had received his doctorate in biology from Harvard in 1958 on the use of electron microscopy to study the frog's optic nerve. He and Lettvin collaborated at the RLE and spent a year developing a way of recording from single unmyelinated optic nerve fibres in the frog. By 1957, they finished, and along with input from McCulloch, Pitts, and Oliver Selfridge, they were ready to publish. This work was highly empirical and required a good deal of experimental drudgery. The 'What the frog's eye tells the frog's brain' paper seemed to define Lettvin's career.

Lettvin's work with Maturana was actually a departure from the original logical approach that McCulloch and Pitts had taken to the brain and perception (McCulloch and Pitts 1943, Pitts and McCulloch 1947). As Lettvin recalled, instead of a model of the mind that involved a system of logical

elements, his work with Maturana suggested that in perception, 'the world was reported by natural language rather than logical language... in terms of things and their relations rather than sense data to be processed into the concepts of things' (Lettvin 1998b, 12). That they discovered that a 'fixed five-layer system of elements, laterally interacting, gave the content of perception to a frog' suggested to Lettvin and Maturana that they were dealing with an analogue, not logical, system (Lettvin 1998b, 12). Here, Lettvin was departing from the logical mechanisms of McCulloch and Pitts to account for perception.

How did McCulloch's and Lettvin's relationship evolve between their time in Chicago and their time at MIT's Research Laboratory of Electronics? In addition to Lettvin coming into his own as an experimentalist at the RLE, with McCulloch more concerned with theoretical questions, their relationship changed too. By the late 1950s, both Pitts and McCulloch were experiencing health problems — Pitt's tragic demise, largely due to Wiener's rejection, has been recounted elsewhere (Smalheiser 2000, Wilson 2010). McCulloch, according to Lettvin, had had a lung episode that when operated on resulted in a heart condition that prevented him from consuming solid food. As this progressed, Lettvin recalled, 'it was clear that [McCulloch] felt that he was decaying in his physical powers'. (Lettvin 1998b, 10). Although McCulloch's enthusiasm for the work at the RLE remained, 'you felt somehow or another you weren't touching him. So Pat and I and Humberto worked more or less independently' (Lettvin 1998b, 10).

There is an interesting contrast between McCulloch's role for Lettvin at Illinois (i.e. during the 1940s) and his role at MIT (1950s). If, during their interrupted time together at the University of Illinois, McCulloch was a typical mentor to Lettvin, by the time they'd arrived at MIT, McCulloch was simply Lettvin's elder collaborator. At Illinois, McCulloch was the leader: writing recommendation letters for Lettvin, running the show. By the time both had arrived at MIT, Lettvin was coming in to his own as a neurophysiologist, and McCulloch took a back seat to the 'youngsters' at the RLE and did much less experimental work. The practices McCulloch engaged in here were connected to the new role he took on.

Throughout their mentor–student relationship during the 1940s as well as their experience as collaborators during the 1950s and 1960s, McCulloch and Lettvin were occupied with the 'materials of science' — the laboratory practices that defined neurophysiology and neurology. However, at the same time, both McCulloch and Lettvin had a self-consciously humanistic approach to the sciences. As I have argued elsewhere, much of McCulloch's work in neurophysiology has been presented by him as a quest to address the problem of mind and brain and to develop a scientific epistemology (Abraham 2003, e.g. McCulloch 1951, McCulloch 1988b), and throughout his career — during his time at Yale during the 1930s and his twilight years at MIT during the 1960s — McCulloch cast the work he was doing on the brain in philosophical terms. Lettvin has described McCulloch as practicing a style of doing and teaching neurophysiology that was unique and one he had never encountered before or since the time they worked closely together (Lettvin 1989a, 332). McCulloch believed that the empirical approach that had long defined studies of the brain and nervous system was simply not adequate to make sense of such a complex system: 'Brute empirics had no excuse for being themselves alone. They had to be revelatory or connected with ideas of one sort or another' (Lettvin 1989a, 332). McCulloch's attitude towards laboratory work and experimentation was complex: although he had

a cavalier attitude, emphasizing instead the importance of ideas and philosophical foundations, he still believed that the nature of cognitive activity lay in structure of the cerebral cortex. McCulloch's was not simply an emphasis on theory at the expense of data. He and Lettvin lamented what for them was a seemingly endless generation of data without any central framework or theory to make sense of it (Lettvin 1989a, 336).

In many ways, both McCulloch and Lettvin were engaged in alternate forms of scientific expression — what Lettvin might have been referring to when he said that McCulloch instilled in him an appreciation for 'the poetry of science'. Poetry and science can intersect in the cultural roles that scientists and poets play, or that science and poetry play, in human existence. But they also align in the ways that science, scientific theories, and scientific practice can be poetic and in the ways that poetry can be a medium through which scientific ideas are expressed. For McCulloch, poetry often served as a mode of autobiographical reflection; for Lettvin, as a mode of playful, irreverent scientific expression. McCulloch, who had been writing poetry since his early days as an undergraduate student at Haverford College, always encouraged students to outdo his fanciful anecdotes, stories, and poems. His vivid imagination stimulated students. In December 1957, McCulloch read a selection of his poems — mostly sonnets — to the Chicago Literary Club. The collection was published as *The Natural Fit* in 1959 (McCulloch 1988a). In the introduction, McCulloch, nearly sixty, alluded to his past as three characters, who, besides having a common interest in 'communication', had little else connecting them (McCulloch 1988a, p. 319). The first was a prospective theologian, who he himself would barely recognize. Here he was referring to his first educational experience at the Quaker Haverford College in Haverford, Pennsylvania. The third, his present self, a scientist. The second 'was born of a war to make the world safe for democracy and was lost in a war without a song' — referring, presumably, to the First World War and the Second World War. It was this character who penned most of poems in the collection, which McCulloch referred to as 'personal communications to his friends'. The last poem, likely written in 1957, was a reflection on his position as a senior scientist:

*farewell sweet morrows hopes deferred and all
crisp years fat earnest in defect of youth
indian summers quicken to keen fall
as brisk october blazons times no ruth*

*i cry no quarter of my age and call
on coming wits to prove the truth
of my start venture into fates cold hall
where thoughts at hazard cast the die for sooth*

*from me great days are gone and after none
array the ardour that i scarce compress
in temperance terrible charged i abide
the desperate victor of my last race run
wanting bold challenge to life's dread excess
to fire that frenzy I must else wise hide*

The second stanza captures McCulloch's role as a mentor during this period. By the late 1950s, it was clear that McCulloch saw that he had accomplished scientifically most of what he was to accomplish in his life, and that he was ready to 'call on coming wits' to continue what he'd begun and 'prove the truth of my start venture into fates cold hall'.

Lettvin heeded McCulloch's call. His stress on the importance of theorizing to gain a proper understanding of the nervous system is shown in Lettvin's translation and adaptation of one of German poet's Christian Morgenstern's (1871–1914) poems, which appeared in the Boston literary magazine *The Fat Abbot* in 1962:

Σ Ξ Man Met a Π Man
After many 'if's and 'but's,
emendations, notes, and cuts,

they bring their theory, complete,
to lay, for Science, at his feet.

But Science, sad to say it, he
seldom heeds the laity

abstractedly he flips his hand,
mutters 'metaphysic' and

bends himself again to start
another curve on another chart

'Come', says Pitts, 'his line is laid;
the only points he'll miss, we've made'.⁴⁶

Clearly referring to his quest, with Pitts and McCulloch, for a coherent theory of brains and nervous systems, Lettvin gestures to the then-dominant attitude towards such pursuits and the divide between scientific and philosophical approaches to the brain, and the endless pursuit of data in the absence of theory. In this way, Lettvin's poetry was at once a form of artistic and scientific expression.

Lettvin as a mentor

By the time they worked as collaborators at MIT, McCulloch had passed on his appreciation for 'the ideas of science' and 'the poetry of science' fully to Lettvin. As McCulloch became more and more occupied with the theoretical aspects of nervous systems, Lettvin was coming into his own as an experimenter while at the same time developing a humanistic approach to the sciences that had been fuelled and encouraged by McCulloch. By 1961, Lettvin had become associate professor at MIT. Although his work at the RLE was primarily experimental, his approach to neurophysiology was always informed by his belief that the traditional separation between philosophy and brain sciences had hampered our understanding of the brain and mind. Lettvin also had a historical perspective as an educator and framed his discussions of brain physiology from a historical point of view. In 1964, Lettvin and de Santillana penned 'A Brazen Proposal for Wedding the Arts and Sciences'.⁴⁷ In this proposal they discussed the need for developing 'scholar-scientists' at MIT. Both those trained strictly in the sciences and those trained in traditional history and philosophy had a disadvantage in properly developing the history and philosophy of science, with scientists unable to appreciate fully the contributions of past researchers and historians and philosophers lacking the experience to 'handle complex scientific ideas and theories with confidence'.⁴⁸ From this vantage point, Lettvin and de Santillana anticipated a strong need for historians and philosophers of science, 'if only

to organize in an understandable fashion the present explosion of knowledge'.⁴⁹ So far this task had been taken on by 'second-raters'. In Lettvin's and de Santillana's view, MIT was in a perfect position to address this need:

First, there is a staff of intelligent men privately very much concerned with principles, hypotheses and the almost inexpressible metaphysical foundations of their work. Second, we have a student body unrivalled in the country, as the tendency of the last two decades has been to push the best young brains in the direction of science and technology.⁵⁰

Lettvin saw the potential for much worthwhile exchange between these groups, perhaps first beginning with about a year of an 'amiable chaos of colloquia' which could become the basis for a new program devoted to the history and philosophy of science, which, if Dean of Humanities Frederick M. Douglas could secure funds, could be guided by de Santillana, Lettvin, and the likes of Noam Chomsky.⁵¹

Lettvin's reportedly well-attended undergraduate lectures from the mid-1960s at MIT also reveal his humanistic scientific perspective, particularly from his course jointly taught with de Santillana, which provided a historical-philosophical perspective on the biological bases of knowledge and perception.⁵² A lecture from January 1965 on the mind-body problem reflects a similar interdisciplinary spirit.⁵³ Rejecting the tendency for the question to be addressed straightforwardly mechanistically in neurophysiological terms, or simply philosophically, Lettvin called for an approach that was integrative, arguing that the

mind-body problem is essentially the relation of neurology to psychology. I have led you a long way around to tell you that there is nothing to tell. There has been no improvement in this regard since Helmholtz. The most important point is that we cannot neglect the philosophy involved. As the positivists would say, we can make gadgets [a gesture to Minsky's AI work], but if they are to really perceive, then we must know what perception is. We cannot make fun of this grappling for words. There is something there and you must begin to talk about what you don't know in order to wrestle it into the known.⁵⁴

Lettvin, like McCulloch, had an irreverent spirit as an educator. One former student remembered Lettvin's neurophysiology classes fondly:

As rigorous as MIT was, you couldn't help but smile as you saw Professor Lettvin's Santa Claus body with hippy-length gray hair bopping down the halls in old-style tennis shoes. He would start by saying 'I am not going to teach this course on what we know about nerves: most of it was known in the '50s, is boring, and would only take a few classes. Instead I am going to teach what we still don't know about nerves. That is what is exciting (Pete A. Sanders Jr '72, in Readers 2006).

Lettvin's wife Maggie recalled of his history of science class: 'He'd go in there and talk for three or four hours and the kids would bring their girlfriends, lunches, and just sit there forever' (Finn 2011).

Lettvin was also a skilled debater, taking on Timothy Leary in 1967 at a student-organized event. Leary had been brought to MIT to defend his argument for the use of LSD: 'Turn off, tune in, drop out'. In his speech following Leary's, Lettvin argued that he saw Leary as a tool of the devil. He made his argument against LSD on moral, not scientific grounds: those who partake in LSD, Lettvin argued, lose a love for the real world, ego, and

experience. It is not like alcohol, or marijuana, where there are reassurances that with the hangover, there is 'judgment to yourself' — when the alcohol or pot wears off, 'you are possibly back to a state where the judgment miraculously is back' (Lettvin 1980, 22). With LSD, Lettvin argued, there was no such reassurance:

I accept pot and I accept whiskey because with either of them there is a guarantee for most people, for almost all people, that you will have another chance, you will come back again and decide whether you want to go back. But it will always be your decision, that is, you will always have freedom once again (Lettvin 1980, 27).

Likening the effects of LSD to epilepsy and temporal lobe lesions, Lettvin argued that what bothered him about LSD was 'the evil of loss, or possible loss of the highest judgmental faculties we have. . . I'm not scared because I will be released to see beautiful pictures. I am scared because of that ultimate me that judges the rest of me, that top noetic function. I don't want to lose it, dammit!' (Lettvin 1980, 27).

Conclusions

McCulloch's relationship with Lettvin shows that his legacy as a cybernetician lies as much with his social and intellectual generosity as with his more straightforward scientific contributions. Although McCulloch's 'laissez-faire' style of mentoring was certainly not unique to him, his relationship with Lettvin and the particular style of scientific practice he fostered in him, says much about cybernetics and mid-twentieth century American science. Ultimately, cybernetics was an effort to transcend disciplinary boundaries, particular with respect to studies of the brain. The cybernetic brain was at once an object of technological, medical, biological, and philosophical interest. This renegotiation of the relations between disciplines and disciplinary boundaries also characterized American science during the World War II and Cold War periods in which cybernetics flourished. Similarly, McCulloch's and Lettvin's 'anarchic' disciplinary attitudes defined their scientific practices, not only in terms of the kinds of questions they asked and their humanistic perspective ('the ideas of science' and 'the poetry of science') but also in their views about the proper study of the brain as requiring both empirical, laboratory-driven practices as well as an overarching theoretical framework ('the materials of science'). Their blurring of humanistic and scientific boundaries, characterized by their poetry and their penchant for history and philosophy of science also can tell us much about what power cybernetics seemed to hold for McCulloch and Lettvin — the development of a coherent theory of the brain and mind.

This essay has relied much on Lettvin's accounts of McCulloch and his experiences of his mentor. What motivated Lettvin to portray McCulloch the way he did? As Laura Otis has shown in her study of Johannes Müller and his students, descriptions of scientists that are based on students' recollections cannot be taken as passive descriptions but as active interpretations that reflect the interests of the student and their understanding of themselves as scientists (Otis 2007). Lettvin's portrayal of McCulloch reflects their very similar academic experiences and scientific perspectives. Both McCulloch and Lettvin got their hands dirty as neurophysiologists, and both initially trained in medicine. Both continued to do laboratory work their whole lives. But

further to this, both had very idiosyncratic ways of asking questions about the brain. Lettvin once said that in his approach to science, he looked for 'the corners that were unoccupied and which, for some reason or another, had escaped attention' (Lettvin 2007). In this sense he was anti-disciplinary — as the questions he asked didn't fit neatly into traditional fields, and in this he was very similar to McCulloch. Lettvin's accounts, then, are not straightforward descriptions but rather a carrying on of McCulloch's scientific and intellectual tradition. This even extended to Lettvin's role as a mentor himself: at the age of seventy-eight, he described the work taking place at the RLE: 'the bibliography of the laboratory is as variegated as it could be, and a source of great pride that I have in my students and postdocs. I say bluntly and sincerely that I served more as a catalyst to their ideas than a source' (Lettvin 1998a, 241).

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Notes

- ¹ Warren S. McCulloch, Interview by Percy Saltzman, 'The Day It Is', Canadian Broadcasting Corporation, c. 1965.
- ² For more on McCulloch's and Lettvin's relationship with Pitts, see Wilson (2009, 2010) and Aizawa (this issue). Wilson argues for the importance of affective relationships in the early history of AI and the homosocial and intergenerational alliances that defined collaborations between McCulloch, Pitts, and Lettvin. My analysis here does not contradict hers; however I focus on McCulloch's relationship with Lettvin in particular.
- ³ I treat McCulloch's intellectual and institutional background in my book *Warren S. McCulloch, Cybernetics, and the Sciences of Brain and Mind, 1930–1970* (MIT Press, forthcoming).
- ⁴ A preliminary list would include, besides Walter Pitts and Jerry Lettvin: Jack Cowan, Humberto Maturana, Patrick Wall, Oliver Selfridge, Marvin Minsky, Seymour Papert, Michael Arbib, Stuart Kauffman, and Heinz von Foerster.
- ⁵ I take this term from Otis's account of 19th-century physiologist Johannes Müller's style of mentoring (Otis 2007, p. 100).
- ⁶ Taffy Holland, interview by Tara Abraham, Old Lyme, Connecticut, May 2001.
- ⁷ Warren S. McCulloch to Jerry Lettvin, 20 August 1942, Folder 'Lettvin, Jerry', Series I, Warren S. McCulloch Papers, B M139, American Philosophical Society Library, Philadelphia, PA; hereafter McCulloch Papers, APSL.
- ⁸ Warren S. McCulloch to Cochrane, 20 August 1942, Folder 'Lettvin, Jerry', Series I, McCulloch Papers, APSL.
- ⁹ Jerry Lettvin to Warren S. McCulloch, c. 1943, Folder 'Lettvin, Jerome Y'. Series I, McCulloch Papers, APSL.
- ¹⁰ Warren S. McCulloch to Arturo Rosenblueth, 29 March 1943, Folder 'Rosenblueth, A.', Series I, McCulloch Papers, APSL.
- ¹¹ Ibid.
- ¹² Walter Pitts to Warren S. McCulloch, 11 September 1944, Series I, McCulloch Papers, APSL.
- ¹³ Denny-Brown's attitude was not surprising — in his position as Director of Harvard's Neurological Unit at BCH, Denny-Brown had a role in building up the discipline of American neurology, particularly in light of neurology's early institutional subsumption under the discipline of psychiatry (Vilensky *et al.* 2004).
- ¹⁴ Jerry Lettvin to Norbert Wiener, ca. April 1946, Box 5.70, Norbert Wiener Papers, MC 22, MIT Archives and Special Collections; hereafter Wiener Papers, MITASC.
- ¹⁵ Ibid.
- ¹⁶ Ibid.
- ¹⁷ Norbert Wiener to Walter Pitts, 4 April 1947, Box 5.77, Wiener Papers, MITASC. Wiener's anger also extended to de Santillana and Selfridge.

- According to McCulloch, Pitts had the manuscript along with other papers in a suitcase in a checking room in New York City, and had given the check to Lettvin who was to pick up the case and bring it to Boston. It remained in New York for several months (McCulloch to Wiener, 8 April 1947 Box 5.77, Wiener Papers, MITASC).
- ¹⁸ Wiener to McCulloch, 5 April 1947, Box 5.77, Wiener Papers, MITASC.
 - ¹⁹ McCulloch to Wiener, 8 April 1947, Box 5.77, Wiener Papers, MITASC.
 - ²⁰ Ibid.
 - ²¹ Ibid.
 - ²² McCulloch to Wiener, 8 April 1947, Box 5.77, Wiener Papers, MITASC.
 - ²³ McCulloch to Lettvin, 15 April 1947, Box 5.77, Wiener Papers, MITASC.
 - ²⁴ Ibid. McCulloch had suspected that Lettvin felt sheepish about a misconceived biochemical hypothesis he'd had and presumably shared with Wiener.
 - ²⁵ McCulloch to Lloyd, 17 September 1947, Folder 'Lloyd, David', Series I, McCulloch Papers, APSL.
 - ²⁶ For more on the use of the stereotaxic instrument in mapping the cerebral cortex, see Abraham (2012).
 - ²⁷ McCulloch to Lloyd, 17 September 1947, Folder 'Lloyd, David', Series I, McCulloch Papers, APSL.
 - ²⁸ Warren S. McCulloch, Recommendation Letter for Jerry Lettvin, n.d. (c. summer 1948), Folder 'Lettvin, Jerome Y.', Series I, McCulloch Papers, APSL.
 - ²⁹ Warren S. McCulloch to David Lloyd, 17 September 1947, Folder 'Lloyd, David', Series I, McCulloch Papers, APSL. See McCulloch and Lettvin (1948).
 - ³⁰ Warren S. McCulloch to Jerry Lettvin, 9 June 1948, Folder 'Lettvin, Jerome Y.', Series I, McCulloch Papers, APSL.
 - ³¹ Warren S. McCulloch to Chief of Division of Research, Graduates and Fellowships, National Research Institute, Bethesda, MD, 24 May 1948, Series I, McCulloch Papers, APSL.
 - ³² Pat Wall recalled that Lettvin took permanent night duty here so that he would work on neurophysiology during the day ('More Data, More Noise: A Celebration of the 60th Birthday of Jerome Y. Lettvin, MIT, February 1980, p. 2; <http://jerrylettvin.blogspot.com/2011/05/more-data-more-noise.html>).
 - ³³ McCulloch had approached Wiener for advice on securing federal funds for Lettvin's work (McCulloch to Wiener, 28 March 1949, Box 6.94, Wiener Papers, MITASC).
 - ³⁴ Jerome Lettvin, Warren S. McCulloch, and Walter H. Pitts, 1948. An Electrical Hypothesis of Central Inhibition and Facilitation. Unpublished Manuscript, Series III, McCulloch Papers, APSL.
 - ³⁵ 'The Research Laboratory of Electronics at the Massachusetts Institute of Technology' December 1945, Box 4.18, Records of the Research Laboratory of Electronics, AC 186, MITASC.
 - ³⁶ McCulloch to Wiesner, 31 May 1951, Folder 'Wiesner, Jerome', Series I, McCulloch Papers, APSL.
 - ³⁷ Walter H. Pitts and Jerry Lettvin to Rosenblueth and Wiener, 14 November 1951, Box 10.144, Wiener Papers, MITASC.
 - ³⁸ H. J. Zimmerman, G. G. Harvey, and W. B. Davenport, Jr, 'Research Review and Objectives of the Research Group in the Research Laboratory of Electronics', Massachusetts Institute of Technology, January 1, 1962, Box 4.45, Records of the Research Laboratory of Electronics, AC 186, MITASC.
 - ³⁹ Wall reports that Wiener's break with McCulloch, Pitts, and Lettvin this time 'was caused by the very generosity and hospitality that was characteristic of McCulloch. Pitts was mortally wounded, seeing in Wiener a father figure, and he was never the same again' (Wall 2001). Conway and Siegelman suggest that Wiener's animosity towards McCulloch and 'the boys' was due to Margaret Wiener's dislike of McCulloch and a false accusation she'd made (Conway and Siegelman 2005, 222–229). Margaret A. Boden cites an explanation offered by McCulloch's student Michael Arbib, who reported that in 1952 Wiener had asked McCulloch to tell him about the brain, and McCulloch's 'mini-lecture had (characteristically) contained as much conjecture as fact' (Boden 2007, pp. 1170–71).
 - ⁴⁰ Norbert Wiener to James R. Killian, 2 December 1951, Box 10.144, Wiener Papers, MITASC.
 - ⁴¹ Ibid.
 - ⁴² Ibid.
 - ⁴³ Norbert Wiener to J. Z. Young, 14 December 1951, Box 10.144. See also Norbert Wiener to Ted Martin, 19 July 1952, Box 10.153, Wiener Papers, MITASC.
 - ⁴⁴ Norbert Wiener to Frank Fremont Smith, 2 April 1953, Box 11.153, Norbert Wiener to Arturo Rosenblueth, 13 November 1953, Box 12.181, Wiener Papers, MITASC.
 - ⁴⁵ Memo from Henry J. Zimmerman to Jerome Y. Lettvin, 6 December 1966, Box 1.1, Jerome Y. Lettvin Papers, MC525, MITASC.
 - ⁴⁶ Lettvin 1962, 8.
 - ⁴⁷ Jerome Y. Lettvin, 'A Brazen Proposal for Wedding the Arts and Sciences', ca. 1964, Box 8.12, Jerome Y. Lettvin Papers, MC 525, MITASC.

That this was Lettvin's initiative, not de Santillana's, is apparent from: Lettvin to Giorgio de Santillana, 9 April 1964, Box 4.6, Jerome Y. Lettvin Papers, MC 525, MITASC.

⁴⁸ Ibid.

⁴⁹ Ibid.

⁵⁰ Ibid.

⁵¹ In 1976, Lettvin became co-director of the Concourse Program for Freshmen at MIT. The Concourse program was founded in 1971 with the aim of creating a learning community for

first-year MIT students that was interdisciplinary and transcended the divide between the sciences and the humanities (MIT News 2011).

⁵² Giorgio de Santillana and Jerry Lettvin, 'Biological Bases of Perception and Knowledge', 21.705, Fall, 1964, Jerome Y. Lettvin Papers, MC 525 Box 2.1, MITASC.

⁵³ Jerome Lettvin, 'The Mind-Body Problem', 5 January 1965, Box 2.1, Jerome Y. Lettvin Papers, MC 525, MITASC.

⁵⁴ Ibid.

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